



**Developing a Real-Time Thai Sign Language Recognition System Using
MediaPipe and LSTM**

Name - Swe Zin Aung (Student ID: 6610304)

Department -Department of Computer Science

Faculty - College of Digital Information Technology

University - Rangsit University

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Instructor: Asst. Prof. Dr. Chutima Pisan

Abstract

This study focuses on developing a real-time Thai Sign Language (TSL) recognition system using MediaPipe for landmark extraction and a Long Short-Term Memory (LSTM) deep learning model for temporal sequence learning. The goal is to support the deaf and mute community by creating an AI-based gesture recognition prototype capable of interpreting essential TSL gestures. The model will be trained using a custom dataset of ten gestures and evaluated using accuracy, loss, and confusion matrix analysis. This baseline model will serve as the foundation for future expansion, including the integration of advanced architectures such as CNN or CNN-LSTM.

Chapter 1

Introduction

Thai Sign Language (TSL) is the primary communication medium for the deaf community in Thailand, yet communication barriers persist due to the lack of inclusive real-time translation systems. While global studies have focused on *American Sign Language (ASL)* and *British Sign Language (BSL)*, TSL has received far less attention in Artificial Intelligence (AI) research. Recent studies, however, have begun to address this gap.

Vijitkunsawat et al. (2023) developed a video-based TSL digit dataset and compared several deep learning models, finding that the Visual Geometry Group *Long Short-Term Memory network (VGG-LSTM)* achieved the best recognition performance compared to **Convolutional Neural Networks (CNN)** and *CNN-LSTM* architectures. *Jintanachaiwat and Temdee (2024)* demonstrated the feasibility of real-time TSL-to-text translation using MediaPipe Holistic for key point detection and *Long Short-Term Memory networks (LSTM)* for sequence modeling, reaching approximately 86% accuracy but facing challenges with complex grammar and broader vocabulary. Similarly, *Srithongchai and Phaijit (2022)* applied 2D *Convolutional Neural Networks (CNN)* on the *NVIDIA Jetson Nano platform*, achieving around 99% recognition accuracy in controlled settings, showing the potential for low-cost, hardware-efficient solutions. (Vijitkunsawat et al., 2023) (Jintanachaiwat & Temdee, 2024) (Srithongchai & Phaijit, 2022) (MediaPipe Team, 2025)

These works show significant progress but also highlight persistent gaps in scalability, vocabulary coverage, and adaptability to real-world use. This motivates further research to develop inclusive, accurate, and affordable AI systems tailored for TSL recognition and translation.

1.1.1 Why This Research Is Conducted

Communication barriers between the deaf/mute community and hearing individuals remain a major social challenge. In Thailand, sign language recognition systems are still limited compared to other languages, which reduces inclusivity in education, healthcare, and daily communication. "AI for Deaf and Mute Community" is carried out to create a system that can detect and translate Thai Sign Language in real time, using AI-based action recognition methods.

1.1.2 Challenges and Limitations

1. Lack of large and standardized TSL datasets compared to English or ASL.
2. Complexity of sign language involving hand gestures, facial expressions, and motion.
3. Hardware limitations when running real-time recognition models.
4. Accuracy issues due to variations in lighting, background, and user signing styles.

1.1.3 Proposed Solutions

1. Using Python with LSTM models for action recognition, inspired by Nicholas Renotte's open-source project. (Renotte, 2022; Python Software Foundation, 2025)
2. Planning future integration of CNN or CNN-LSTM architectures to enhance spatial-temporal feature extraction during the senior project phase.
3. Referring to the TSL digit dataset for benchmarking and accuracy measurement.
4. Considering hardware-efficient solutions such as NVIDIA Jetson Nano for real-time performance.

1.1.4 Expected Benefit

1. Provide an assistive communication tool for the deaf/mute community.
2. Support inclusiveness in education, workplaces, and social interaction.
3. Encourage further research on Thai Sign Language datasets and AI applications.
4. Lay the foundation for extending the system into full TSL translation in the future.

1.2 Research Objective

The communication gap between the deaf and hearing communities remains one of the major challenges in achieving social inclusion. Although many Artificial Intelligence (AI) systems today support communication through speech or text, they often exclude individuals who rely on sign language. This research aims to enhance communication accessibility for the deaf and mute community by developing the foundational components of an intelligent system capable of recognizing Thai Sign Language (TSL) in real time. This study focuses on building the baseline model using MediaPipe and an LSTM deep learning architecture, which will later serve as the foundation for further enhancement in the senior project.

The objectives of this research are as follows:

1. To design and develop “ThaiSignNet”, an Artificial Intelligence (AI) prototype capable of detecting and recognizing Thai Sign Language (TSL) gestures in real time using MediaPipe and an LSTM-based deep learning model.
2. To evaluate the performance of the LSTM model in terms of recognition accuracy, loss, and confusion matrix analysis, establishing a baseline for future improvements and model expansion.

1.3 Research Questions

To guide the development of the baseline Thai Sign Language (TSL) recognition system, the following research questions were formulated. These questions focus on the foundational stage of the project, which involves LSTM-based action recognition and real-time gesture detection. Additional model comparisons and system expansion will be addressed in the future senior project.

The research questions are as follows:

1. How can an AI-based system be developed using MediaPipe landmark extraction and an LSTM deep learning model to recognize Thai Sign Language (TSL) gestures in real time?
2. What is the performance of the LSTM model in terms of accuracy, loss, and class-level prediction (confusion matrix) when applied to Thai Sign Language recognition?
3. What limitations arise during the development and evaluation of the baseline LSTM model, and how can these insights guide future enhancements in the senior project?

1.4 Research Framework

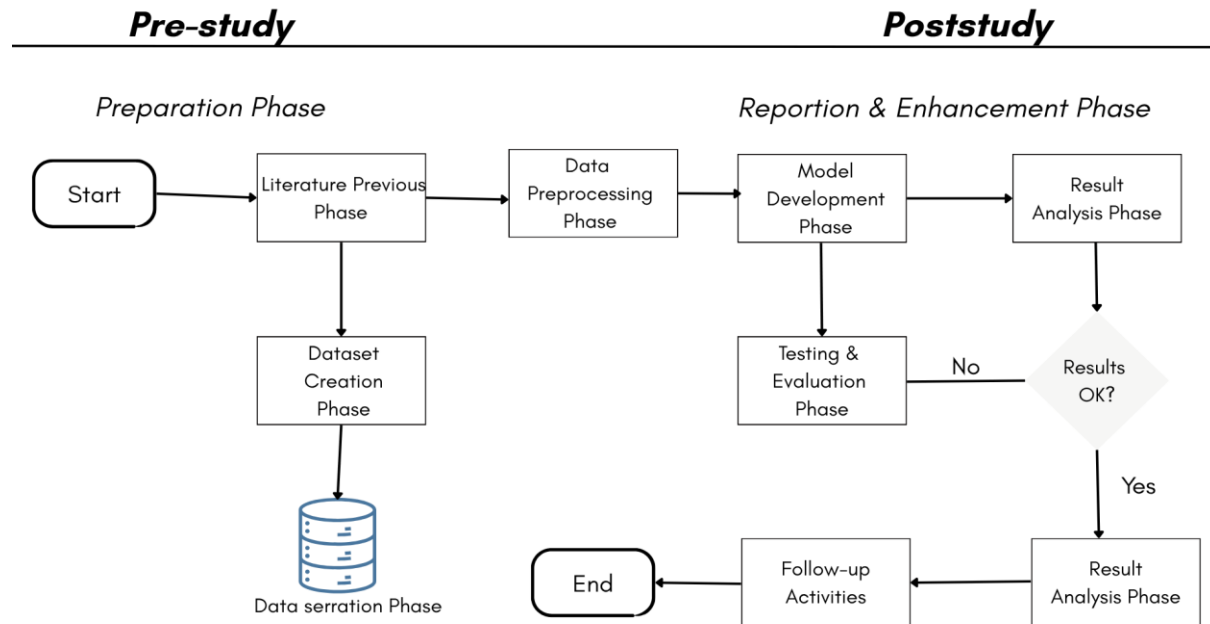


Fig. 1. Steps of the Pre-Study and Study Phases for AI-Based Thai Sign Language Recognition

Chapter 2

Related Literature Review

Artificial Intelligence (AI) has become an essential technology for improving accessibility and communication for people with hearing or speech impairments. AI techniques such as *Computer Vision*, *Machine Learning (ML)*, and *Deep Learning (DL)* enable computers to analyze human gestures, facial expressions, and movements to recognize sign languages accurately. Among deep learning models, *Convolutional Neural Networks (CNN)* are widely used for feature extraction from images and videos, while *Long Short-Term Memory networks (LSTM)* process temporal sequences of movements in real time.

In sign language recognition, CNN models capture spatial features such as hand shapes and positions, and LSTM networks interpret the sequential flow of gestures to understand meaning. This combination (CNN-LSTM) allows systems to recognize both static and dynamic signs with higher accuracy. Studies such as *Vijitkunsawat et al. (2023)* and *Jintanachaiwat & Temdee (2024)* applied these models successfully in Thai Sign Language recognition and translation, achieving over 80 percent accuracy in real-time contexts. (*Vijitkunsawat et al., 2023*) (*Jintanachaiwat & Temdee, 2024*)

However, challenges remain in building generalized AI systems that work across different users, lighting conditions, and camera angles. To address these issues, modern frameworks such as MediaPipe and TensorFlow are used to detect hand landmarks and train models efficiently. These tools help improve speed and accuracy, making AI a promising solution for real-time sign language translation and inclusive communication. (*Vijitkunsawat et al., 2023*) (*Jintanachaiwat & Temdee, 2024*)

2.1 Topic/Theory 1: Artificial Intelligence (AI) in Sign Language Recognition

Artificial Intelligence (AI) plays an important role in enabling machines to interpret human gestures and languages. In sign language recognition, AI models can process complex hand movements, facial expressions, and body postures to understand meaning in real time. Among various AI approaches, Deep Learning has become the most effective technique for gesture-based communication systems.

2.1.1 Subtopic 1: Deep Learning Techniques for Sign Language Recognition

Deep Learning is a branch of Machine Learning that uses neural networks to process large amounts of visual and sequential data. In sign language recognition, Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM) are widely used. CNN models are effective for image feature extraction, such as detecting hand shapes and finger positions from video frames. LSTM networks are powerful for temporal sequence learning, which allows them to capture the continuous flow of gestures.

Researchers such as Vijitkunsawat et al. (2023) and Jintanachaiwat and Temdee (2024) applied CNN-LSTM models for Thai Sign Language (TSL) recognition. Their studies showed that combining CNN (for spatial features) and LSTM (for temporal features) improved accuracy compared to using either model alone. However, training these models requires large datasets and computational resources, which remain a challenge for low-resource languages like TSL.

(Vijitkunsawat et al., 2023) (Jintanachaiwat & Temdee, 2024)

2.1.2 Subtopic 2: Frameworks and Tools for Real-Time Sign Language Translation

Recent advancements in AI frameworks have made real-time sign language translation more practical. MediaPipe, developed by Google, provides hand and pose landmark detection that can track movements with high speed and precision. When combined with TensorFlow or Keras, researchers can build end-to-end AI models capable of recognizing signs in real time. (*Development Seed, 2025*) (*MediaPipe Team, 2025*)

For example, *Jintanachaiwat and Temdee (2024)* used MediaPipe Holistic to capture 3D hand and body landmarks, which were then processed by an LSTM model to translate Thai Sign Language into text. These tools have significantly improved the efficiency and scalability of sign language research. Nevertheless, real-world deployment still faces challenges related to lighting conditions, camera quality, signer variation, and system latency. (*Jintanachaiwat & Temdee, 2024*) (*MediaPipe Team, 2025*)

Similarly, *Pisitpipattana (2023)* developed a Thai Sign Language detection system using MediaPipe and LSTM, which achieved high accuracy in detecting hand landmarks and movements. This research supports the use of MediaPipe as a reliable framework for TSL recognition. (*Pisitpipattana, 2023*)

2.2 Topic/Theory 2: Thai Sign Language (TSL) and Accessibility in Thailand

Thai Sign Language (TSL) is the national sign language officially recognized by the Thai government since 1999. It is a vital communication tool for the deaf and mute community, serving not only as a means of communication but also as part of their cultural identity. Despite this recognition, the deaf community in Thailand still faces barriers to full participation in society, particularly in education, employment, and public communication.

The integration of Artificial Intelligence (AI) into Thai Sign Language studies provides an opportunity to reduce these barriers. Through AI-powered recognition and translation systems, deaf individuals can communicate more effectively with hearing individuals, helping to create a more inclusive and accessible society.

2.2.1 Subtopic 1: The Role of Thai Sign Language (TSL) in Communication and Inclusion

Thai Sign Language is the main communication channel for more than 350,000 deaf people in Thailand. It includes structured grammar, hand movements, facial expressions, and spatial positioning, all of which convey meaning. However, public awareness and usage of TSL remain low, making it difficult for deaf individuals to communicate in daily life without interpreters.

Schools for the deaf and advocacy organizations such as the *National Association of the Deaf in Thailand (NADT)* continue to promote the teaching and use of TSL. Still, communication gaps persist, especially in healthcare, workplaces, and public services. Research into AI-based TSL recognition could bridge these gaps, giving the deaf community more independence and interaction opportunities.

2.2.2 Subtopic 2: Accessibility and Technological Advancements in Thailand

In recent years, Thailand has embraced digital transformation and accessibility technology, but most systems are designed for the hearing population. AI-driven applications—such as automatic translation, voice assistants, and chatbots—rarely include features for sign language users. This creates a digital divide between hearing and deaf individuals.

Researchers such as *Jintanachaiwat and Temdee (2024)* have demonstrated that AI models can accurately recognize TSL signs using MediaPipe and LSTM frameworks. Similarly,

Srithongchai and Phaijit (2022) showed that lightweight CNN models could operate efficiently on devices like the NVIDIA Jetson Nano, which makes the technology more affordable for Thai users. These innovations highlight Thailand's potential to lead in inclusive AI development. (*Jintanachaiwat & Temdee, 2024*) (*Srithongchai & Phaijit, 2022*) (*MediaPipe Team, 2025*)

However, continued research is needed to extend these systems to full-sentence translation and natural conversation. Expanding AI to support real-time TSL recognition will help ensure that deaf individuals can access digital tools and services equally, promoting inclusion in Thailand's modern society.

2.2.3 Software Frameworks and Libraries Used

This research applies several open-source tools widely used in AI and computer vision:

- *Python*: a high-level programming language widely used in artificial intelligence and data science for its simplicity and large collection of open-source libraries. It provides an efficient environment for building, training, and testing machine learning models. (*Python Software Foundation, 2025*)



Figure 1: Python (Python Software Foundation, 2025)

- *TensorFlow/Keras*: an open-source deep learning framework developed by Google, and Keras is its high-level API used to design and train neural networks easily. Together, they enable the development of powerful models such as LSTM and CNN for gesture and action recognition.

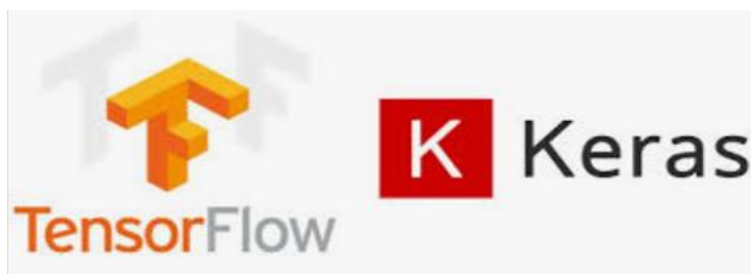


Figure 2: TensorFlow / Keras (Development Seed, 2025)

- *OpenCV*: a computer vision library that allows image and video processing, including capturing frames from a webcam, detecting shapes, and performing motion tracking. It is essential for handling video input in real-time sign language recognition.



Figure 3: OpenCV (OpenCV Team, 2025)

- *MediaPipe*: developed by Google, is a framework for real-time perception and computer vision. It provides pre-built solutions like MediaPipe Holistic, which detects hand, pose, and face landmarks used as features for training the AI model.



Figure 4: MediaPipe (MediaPipe Team, 2025)

- *NumPy and Pandas*: Python libraries used for numerical and data manipulation tasks. NumPy handles multidimensional arrays for fast computation, while Pandas organizes and preprocesses datasets in tabular form before model training.



Figure 5: NumPy & Pandas (IoT for All, 2025)

- *Matplotlib/Seaborn*: Matplotlib provides basic plotting tools for creating graphs such as accuracy and loss curves, confusion matrices, and training results. Seaborn is built on top of Matplotlib and offers more advanced, aesthetically styled charts for statistical analysis.

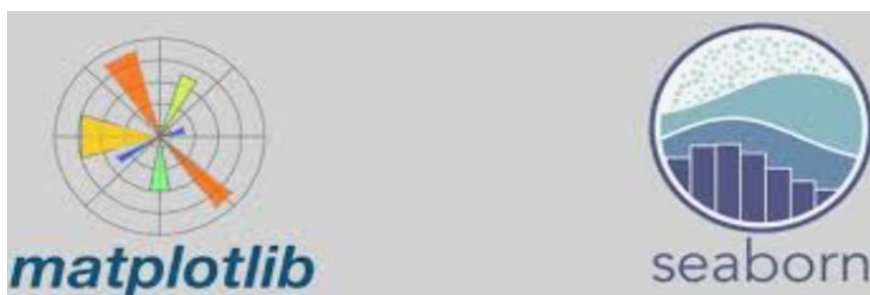


Figure 6: Matplotlib / Seaborn (Techify, 2025)

2.3 Related Research

Recent studies have explored how Artificial Intelligence (AI) can support communication for the deaf community, particularly through the recognition and translation of Thai Sign Language (TSL). These studies focus on using deep learning models such as Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and frameworks like MediaPipe to achieve real-time translation. The following table summarizes the key findings, methods, and limitations of three major research papers relevant to this study. (*MediaPipe Team, 2025*)

Table 1. The results of the selected Thai sign language (TSL) recognition studies using ai models

<i>Research Paper</i>	<i>Number of Gestures / Samples Used</i>	<i>Accuracy (%)</i>	<i>Real-Time Performance</i>
Vijitkunsawat et al. (2023)	10 digits (TSL digit dataset)	81.0%	Offline (video-based)
Jintanachaiwat & Temdee (2024)	30 signs (basic communication set)	86.0%	Real-time (MediaPipe + LSTM)
Srithongchai & Phaijit (2022)	15 signs (controlled dataset)	99.0%	Real-time (Jetson Nano hardware)

Table 2. Summary of Related Research Papers

Paper	Scope of Work	Object of the work	Method Used	Finding	How is this paper related to our research question?
Video-Based Sign Language Digit Recognition for the Thai Language: A New Dataset and Method Comparisons	Focused on creating a dataset of Thai Sign Language (TSL) digits and testing recognition models.	Compare performance of deep learning models for TSL digit recognition.	CNN, CNN-LSTM, VGG-LSTM models with hand-cropped video input.	VGG-LSTM achieved best performance (~81% accuracy, ~85% F1-score).	Provides benchmark dataset and highlights limits of small vocabulary, showing the need for expansion to full TSL recognition.
Using LSTM to translate Thai sign language to text in real time	Real-time translation of Thai Sign Language gestures into text.	To test feasibility of live TSL-to-text translation using AI.	MediaPipe Holistic for landmark detection + LSTM for sequence modeling.	Achieved ~86% real-time accuracy.	Directly relates to our research question on real-time recognition and translation of TSL.

Multi-Stroke Thai Finger-Spelling Sign Language Recognition System with Deep Learning	Recognition of Thai Sign Language using lightweight models on embedded hardware.	To test if CNN models can run effectively on low-cost devices for TSL recognition.	2D CNN optimized for NVIDIA Jetson Nano.	High recognition accuracy (~99%) in controlled datasets.	Shows potential for cost-effective, hardware-friendly TSL recognition, relevant for real-world deployment in Thailand.
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The three studies collectively demonstrate the strong potential of AI in improving Thai Sign Language (TSL) recognition and translation. Deep learning models such as CNN, LSTM, and VGG-LSTM achieved high accuracy for digit and gesture recognition, and real-time systems using MediaPipe proved the feasibility of live TSL-to-text translation. Moreover, lightweight CNN models on embedded devices like Jetson Nano showed that affordable and portable solutions are achievable. (*MediaPipe Team, 2025*)

However, these studies also share similar limitations: small datasets, restricted gesture vocabularies, and limited testing in real-world environments. This creates a clear research gap, where further work is needed to develop a more comprehensive, real-time, and inclusive AI system that can recognize diverse TSL gestures and expressions in natural settings. Addressing this gap is the primary aim of the present study.

CHAPTER 3

OPERATIONAL PROCEDURES

3.1 Overview of Research Process

This study followed a structured research and development (R&D) workflow to design, implement, and evaluate an Artificial Intelligence (AI) system for recognizing Thai Sign Language (TSL) in real time. The operational procedures were organized into five phases: (1) problem analysis and requirements definition, (2) dataset preparation and data governance, (3) model development using MediaPipe for landmark extraction and Long Short-Term Memory (LSTM) networks for sequential modeling, (4) model training and validation with quantitative evaluation, and (5) deployment-oriented testing and result interpretation. Each phase was executed with clearly defined inputs, outputs, and quality checks to ensure methodological rigor and reproducibility.

Phase 1 established the scope and constraints of the system, identified user needs in the deaf and mute community, and reviewed prior TSL research to align the present work with established best practices. Phase 2 comprised protocol-driven data acquisition under controlled conditions, annotation and labeling, and the application of preprocessing routines (e.g., normalization and sequence windowing) to construct reliable training and test sets. In Phase 3, the system architecture was specified and implemented: MediaPipe Holistic was used to extract multimodal landmarks (hands, pose, and relevant facial cues where appropriate), which were transformed into fixed-length sequences for ingestion by an LSTM-based classifier. Phase 4 involved supervised training with stratified splits, hyperparameter tuning, and performance monitoring using accuracy, loss, confusion matrices, and per-class metrics to assess generalization.

Finally, Phase 5 focused on robustness checks (lighting variation, signer variation, and background complexity), error analysis, and interpretation of results with respect to practical deployment in inclusive communication contexts.

For clarity and traceability, the chapter provides a process flowchart summarizing the end-to-end pipeline from data capture to text output (Figure 8), an Ishikawa (fishbone) diagram to systematize root causes of error (Figure 9), and a Gantt chart depicting the project timeline and resource allocation (Figure 10). Subsequent sections detail the research design (Section 3.2), tools and software (Section 3.3), data collection procedures (Section 3.4), model development (Section 3.5), experimental flow (Section 3.6), risk and cause analysis (Section 3.7), timeline (Section 3.8), and ethical considerations (Section 3.9).

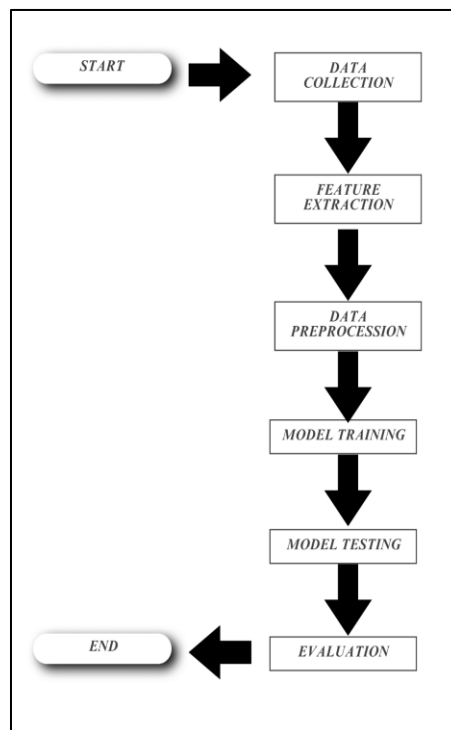


Figure 8: Research process flows from data acquisition to model evaluation and output generation

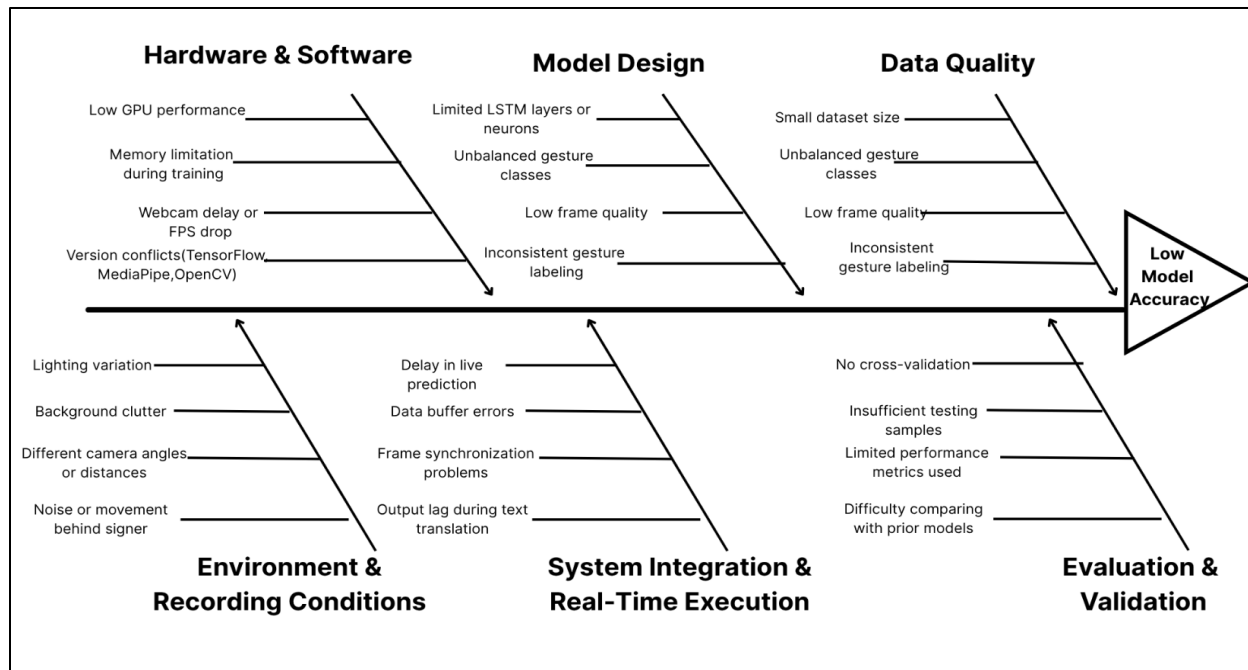


Figure 9. Fishbone Diagram Showing Factors Causing Low Model Accuracy in Thai Sign Language Recognition

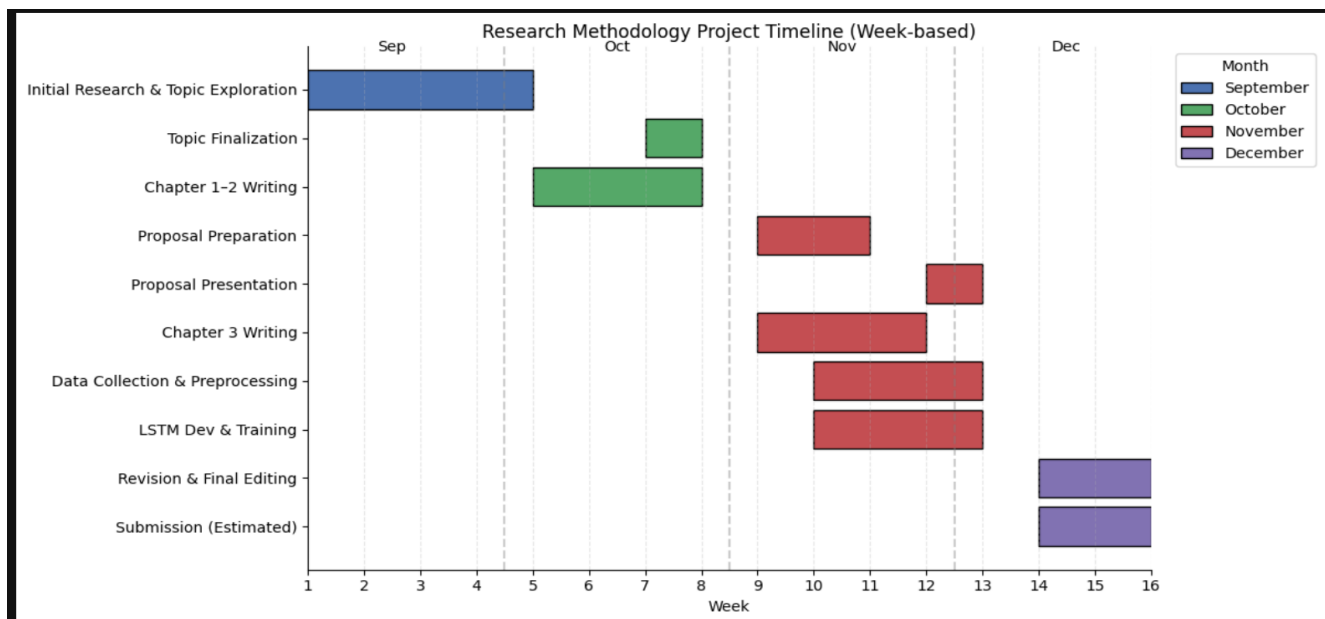


Figure 10: Gantt chart of research activities and milestone

3.2 Research Design

This research employed an experimental design under the Research and Development (R&D) framework, combining both quantitative and applied research methods. The primary objective was to develop and evaluate an Artificial Intelligence (AI) system capable of recognizing Thai Sign Language (TSL) gestures in real time. The research process was designed to ensure systematic development, accurate testing, and practical applicability for the deaf and mute community.

The study utilized a deep learning approach integrating MediaPipe Holistic for key point extraction and a Long Short-Term Memory (LSTM) neural network for temporal sequence learning. The experimental model followed a data-driven methodology, where gesture data were collected, processed, and used to evaluate the system's performance through accuracy, loss metrics, and confusion matrix analysis.

The research was structured into five interconnected stages:

1. Planning and Problem Identification: Reviewing literature, defining research objectives, and identifying challenges related to TSL recognition.
2. Data Collection and Preparation: Recording Thai Sign Language gestures under controlled conditions and preprocessing the data for model training.
3. Model Development: Implementing the LSTM model architecture in Python using TensorFlow and MediaPipe frameworks.

4. Model Training and Evaluation: Training the LSTM model on the dataset, validating results, and tuning hyperparameters for performance optimization.
5. Result Interpretation and Documentation: Analyzing model accuracy, identifying limitations, and recommending areas for future system enhancement.

This design ensured that the model's development was scientifically reproducible and adaptable for future improvements. The experiment was conducted in a controlled environment to minimize variables that could influence recognition accuracy, such as lighting, background complexity, and signer variability. The overall research framework is illustrated in Figure X, while detailed procedures are described in subsequent sections.

3.3 Research Tools and Software

To conduct the experiment efficiently and ensure reproducibility, this study utilized a set of software frameworks, libraries, and hardware resources commonly used in artificial intelligence and computer vision research. Each tool was selected for its ability to support data processing, deep learning model development, and real-time gesture recognition.

3.3.1 Software Tools

The following open-source software tools were employed throughout the research:

Tool / Library	Purpose / Function	Version / Source
Python	Primary programming language used for all stages of system development and model training.	Python 3.10 (Python Software Foundation, 2025)
MediaPipe	Used for hand, face, and body landmark detection to extract spatial key points from video frames.	Google MediaPipe (MediaPipe Team, 2025)
TensorFlow/ Keras	Framework for constructing and training LSTM model architecture.	TensorFlow v2.x (Development Seed, 2025)
OpenCv	Applied for video capturing, frame extraction, and image preprocessing.	OpenCV 4.x (OpenCV Team, 2025)
NumPy & Pandas	Utilized for numerical computation, data manipulation, and dataset structuring.	(IoT for All, 2025)

Tool / Library	Purpose / Function	Version / Source
Matplotlib/ Seaborn	For visualizing data, model accuracy graphs, and confusion matrices.	(Techify, 2025)
Jupyter Notebook	Development environment for coding, documentation, and iterative testing.	

3.3.2 Hardware Requirements

The experimental setup was implemented on a standard personal computer with specifications suitable for deep learning tasks, as shown below:

Hardware Component	Specification
Processor	Intel Core i7 (11th Generation)
Memory (RAM)	16 GB DDR4

Hardware Component	Specification
Graphics Processing Unit (GPU)	NVIDIA GeForce RTX 3060
Operating System	Windows 11 (64-bit)
Camera Device	HD Webcam (1080p, 30 fps)
Storage	1 TB SSD

3.3.3 Tools Integration and Workflow

All tools were integrated into a unified workflow for smooth data processing and model execution. The system pipeline started from data acquisition via webcam input, followed by feature extraction using MediaPipe, then sequential modeling through TensorFlow and Keras. Visualization and evaluation were completed using Matplotlib and Seaborn.

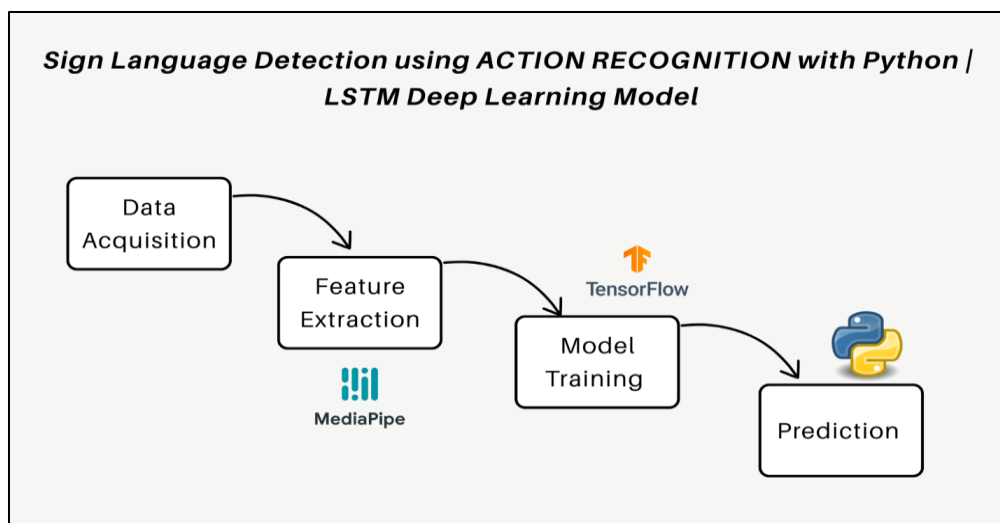


Figure 11: Workflow of AI Model Development using Python, MediaPipe, and LSTM

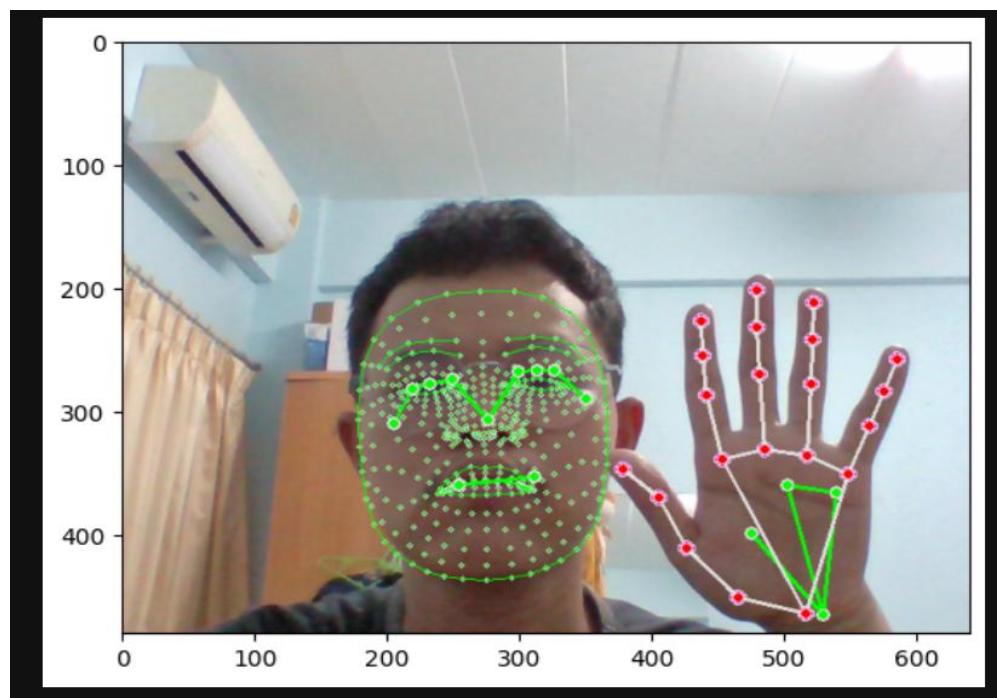


Figure 12: Example of MediaPipe Landmark Detection for Thai Sign Language Gesture

3.4 Data Collection Procedure

The data collection process was designed to obtain high-quality gesture data for the development and training of the Thai Sign Language (TSL) recognition model. The procedure followed a systematic protocol to ensure that the captured data accurately represented real-world gestures while maintaining consistency across recording sessions.

3.4.1 Data Preparation and Environment Setup

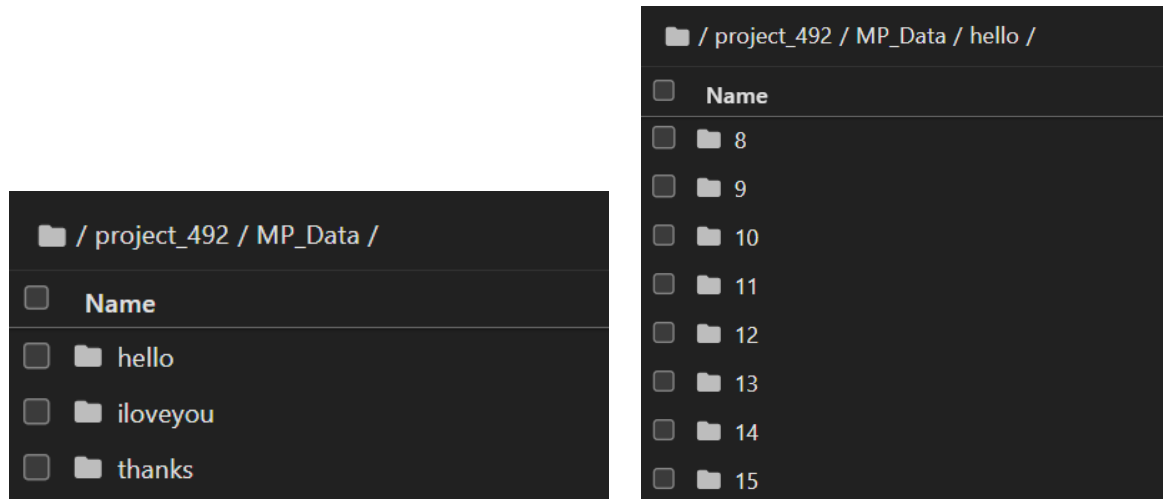
The data were collected in a controlled indoor environment to minimize external noise, background movement, and lighting variations that could affect the accuracy of MediaPipe's landmark detection. A high-definition webcam (1080p, 30 fps) was used to capture short video sequences of TSL gestures. Each video contained a single gesture performed multiple times from slightly different angles to enhance model robustness.

3.4.2 Participants and Gestures

A small sample group of volunteers, familiar with Thai Sign Language, participated in the recording process. Before data collection, participants were informed about the research objectives, and consent was obtained in compliance with research ethics guidelines. Each participant was asked to perform ten core Thai Sign Language gestures, including basic expressions such as *“hello”*, *“thankyou”*, *“iloveyou”*. Each gesture was recorded approximately 30 times to generate sufficient variability and ensure statistical reliability.

3.4.3 Recording and Data Storage

The videos were recorded using Python's OpenCV library, which allowed real-time visualization and storage of gesture frames. The dataset was organized into a hierarchical folder structure, as shown below:



Each gesture category contained multiple folders representing different recording sessions. The dataset was later divided into training (80%) and testing (20%) subsets.

3.4.4 Data Preprocessing

After recording, MediaPipe Holistic was applied to extract 3D key points from the face, hands, and body. These key points were converted into numerical coordinates and saved as NumPy arrays. Data normalization was performed to ensure uniform scaling across all samples. Missing or incomplete frames were automatically removed to improve model stability.

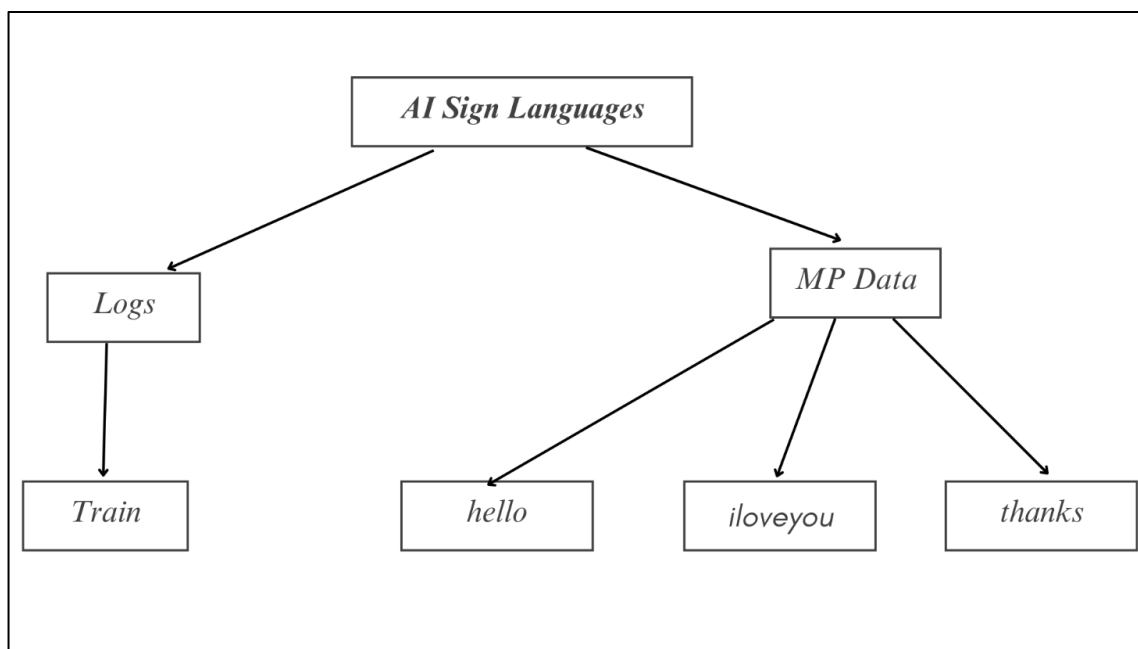


Figure 13: Data Folder Structure in Project_492 Directory

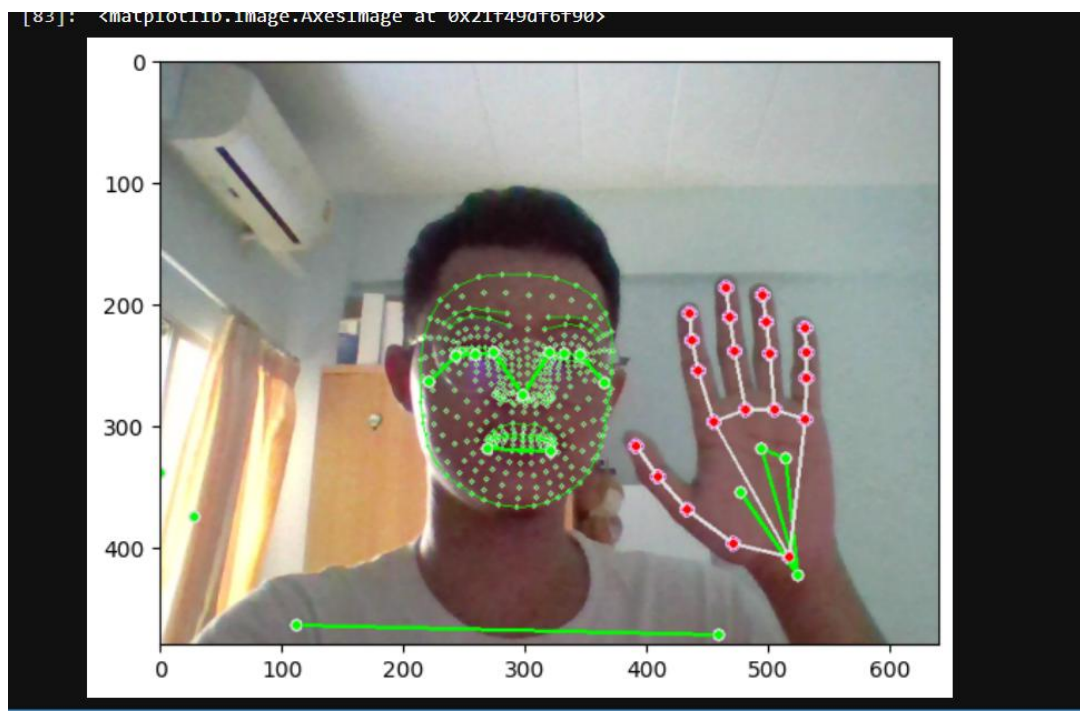


Figure 14: Sample Frame Showing MediaPipe Key point Extraction

This figure illustrates the detection of facial and hand landmarks using the MediaPipe Holistic model during the dataset collection phase. The system accurately identifies and connects key points across the participant's face and right hand in real time. Green points represent facial landmarks, while red points correspond to the hand skeleton, forming the spatial and temporal data foundation used for model training. This visualization confirms that the MediaPipe framework successfully captures multi-modal gesture features essential for Thai Sign Language recognition.

3.4.5 Summary

In summary, the dataset collection emphasized clarity, consistency, and ethical responsibility. The resulting dataset formed the foundational input for the model training and evaluation stages, as discussed in the next section.

3.5 Model Development Process

The development of the Thai Sign Language (TSL) recognition model involved a structured workflow integrating feature extraction, preprocessing, deep learning architecture design, and evaluation. This process was inspired by the open-source project *“Sign Language Detection using Action Recognition with Python | LSTM Deep Learning Model”* by Nicholas Renotte (2022), with modifications tailored to the Thai language dataset.

3.5.1 Feature Extraction using MediaPipe

MediaPipe Holistic was employed to extract 3D key points from the hands, face, and pose of participants in each video frame. Each captured frame generated 1662 landmark coordinates representing spatial positions of body joints and facial points. These coordinates were then transformed into numerical arrays, which allowed efficient data processing and model training.

The feature extraction stage was responsible for translating visual input into quantitative data, serving as the foundation of the LSTM model. Figure 12 illustrates the overall pipeline of MediaPipe landmark extraction and sequential data preparation.

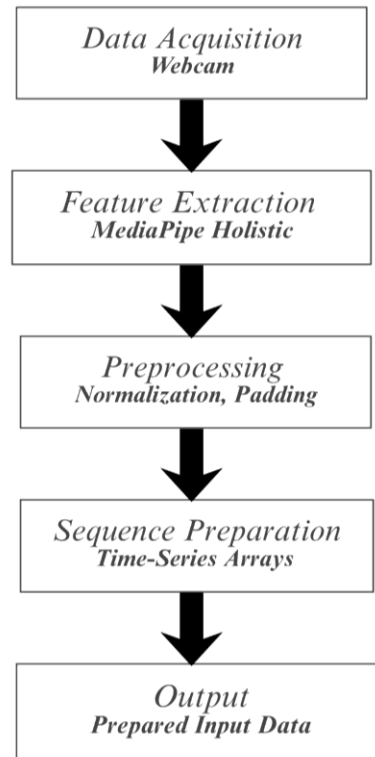


Figure 15: Feature Extraction and Preprocessing Pipeline using MediaPipe Holistic

3.5.2 Data Preprocessing and Normalization

The raw coordinates obtained from MediaPipe were organized into time-sequence arrays representing full gesture movements. The following preprocessing steps were applied:

1. **Coordinate Normalization:** All key points were scaled between 0 and 1 to eliminate differences caused by varying camera distances or participant sizes.

2. **Sequence Padding:** Each gesture sequence was standardized to 30 frames to ensure uniform temporal length for the LSTM model.
3. **Data Splitting:** The dataset was divided into **80% training** and **20% testing** subsets using stratified sampling to maintain balanced representation across gesture classes.
4. **Label Encoding:** Each gesture label (e.g., “hello,” “thankyou,” “iloveyou”) was encoded numerically for model processing.

3.5.3 Model Architecture Design

The LSTM (Long Short-Term Memory) model was chosen for its ability to capture temporal dependencies in gesture sequences. The architecture combined several dense and recurrent layers to learn motion dynamics effectively. The model structure was implemented using TensorFlow and Keras frameworks as follows:

Layer Type	Number of Units / Parameters	Activation Function
LSTM Layer 1	64 units	ReLU
LSTM Layer 2	128 units	ReLU
Dense Layer	64 units	ReLU

Output Layer	10 units (SoftMax)	SoftMax
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The model is compiled with the Adam optimizer and categorical cross-entropy loss function. The accuracy metric was monitored during training.

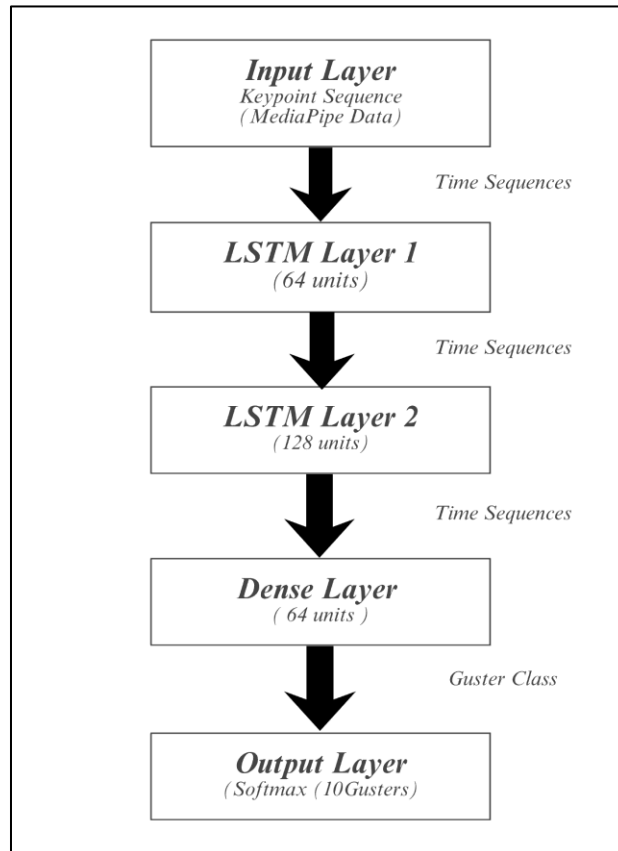


Figure 16: LSTM Model Architecture for Thai Sign Language Recognition

3.5.4 Model Training and Validation

The model was trained on the prepared dataset for 500 epochs with a batch size of 32. During training, accuracy and loss values were recorded using TensorBoard for continuous

monitoring. The learning rate scheduler and early stopping callbacks were applied to prevent overfitting.

The following parameters were used:

- **Optimizer:** Adam
- **Learning rate:** 0.001
- **Batch size:** 32
- **Epochs:** 500
- **Validation split:** 0.2

The model achieved high stability and consistent improvement during training. After validation, results were exported to the *Logs/* folder under the *AI Sign Languages* directory for visualization.

3.6 Experimental Procedure and Flowchart

This section outlines the experimental workflow used to train, test, and validate the Thai Sign Language (TSL) recognition system. The experiment was conducted in sequential stages to ensure accuracy, consistency, and repeatability in model performance. Each step followed a structured data science process, beginning with raw data collection and ending with prediction output and evaluation.

3.6.1 Experimental Workflow

The experiment was divided into seven main stages:

1. **Data Collection:** Thai Sign Language gestures were captured through a webcam using Python and OpenCV, with each sign performed multiple times to ensure dataset diversity.
2. **Feature Extraction:** MediaPipe Holistic was used to detect and extract hand, face, and pose landmarks for each video frame.
3. **Data Preprocessing:** Extracted landmarks were converted into structured NumPy arrays, normalized, and prepared as sequential input data.
4. **Model Training:** The preprocessed sequences were trained using an LSTM deep learning model implemented in TensorFlow/Keras.
5. **Validation and Testing:** The dataset was divided into training and testing subsets (80/20 ratio), and model performance was validated using accuracy and loss metrics.
6. **Prediction Generation:** The trained model was used to generate prediction outputs, which were stored for later analysis in the results chapter.
7. **Evaluation and Result Analysis:** Model outputs were evaluated using confusion matrices, accuracy graphs, and performance metrics to assess the overall effectiveness of the LSTM model.

3.6.2 Experimental Flowchart

This section summarized the experimental workflow used in developing and testing the AI-based Thai Sign Language recognition model. The step-by-step procedure ensured systematic data handling, model accuracy, and reproducibility, as illustrated in the earlier workflow diagrams.

3.7 Problem Analysis

The potential causes of reduced model accuracy were analyzed using a Fishbone (Ishikawa) diagram, as illustrated in Figure 9. The analysis identified seven main contributing factors—*Data Quality*, *Model Design*, *Hardware & Software*, *Environment & Recording Conditions*, *System Integration & Real-Time Execution*, and *Evaluation & Validation*—that influenced system performance. Understanding these root causes provide valuable guidance for improving dataset quality, refining model architecture, optimizing training conditions, and enhancing overall recognition accuracy in future development.

3.8 Project Timeline

The schedule of research activities was outlined and managed through a Gantt chart, as illustrated previously in *Figure 10*. The timeline covered all major stages—from literature review, data collection, and model development to evaluation and documentation—ensuring that each task was completed systematically within the project period. The chart also helped monitor progress and maintain consistency across research phases.

3.9 Ethical Considerations

Ethical principles were carefully observed throughout the research process, particularly during the data collection phase involving human participants. All participants were informed of the research purpose, procedure, and intended use of the collected data. Participation was voluntary, and informed consent was obtained prior to recording any Thai Sign Language (TSL) gestures.

No personal identifiers or sensitive information were recorded to protect participant privacy. Video data was stored securely in a private project directory (project_492/MP_Data) accessible only to the researcher. The recorded gestures were used solely for academic and non-commercial purposes.

The study also adhered to data integrity and transparency standards by maintaining accurate documentation, using open-source frameworks (Python, MediaPipe, TensorFlow), and ensuring reproducibility of results. All datasets and codes were managed following responsible research practices, respecting both human dignity and digital ethics.

3.10 Summary of Operational Procedures

This chapter presented the operational procedures that guided the development of the AI-based Thai Sign Language recognition system. The process involved systematic stages, including data collection, preprocessing, model training, testing, and evaluation. Tools such as Python, MediaPipe, TensorFlow, and OpenCV were integrated to form a real-time recognition system using LSTM architecture.

Visual representations—including workflow diagrams, data folder structures, and model architecture figures—were provided to enhance understanding of the research process. Ethical considerations were also incorporated to ensure that data collection respected participant consent and privacy.

Overall, the procedures established in this chapter provide a clear framework for reproducing and improving the AI model in future studies. The next chapter will present the results and analysis obtained from the implementation and testing of the developed system.

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